

Baron Eiley

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Professional Experience

Lockheed Martin Space – Sunnyvale, California

August 2022 – Present

FPGA RTL Design and Architecture

- Developed SystemVerilog RTL for long-range, high-altitude vehicles to control the arming and disarming of critical hardware functions. Designs included robust finite state machines and logic to process real-time accelerometer data from ADCs.
- Designed Simulink-based models to generate synthesizable RTL signal processing modules, including interpolators, decimators, and cascaded integrator-comb (CIC) filters.
- Validated and debugged signed fixed-point RTL for arithmetic operations processing accelerometer data using SystemVerilog test-benches, ensuring accurate FPGA-based mathematical integration to produce velocity and distance values for high-altitude interceptor vehicles.
- Architected, synthesized, and maintained FPGA-based top-level designs integrating algorithmic modules with System-on-Chip (SoC) hard processors using AMD Vivado and Microchip Libero. Designs targeted AMD Zynq Ultrascale+ and Microchip RTG4/PolarFire devices, utilizing high-performance bus protocols such as AXI and APB.
- Continuously synthesized and delivered updated FPGA bitstreams to the algorithms team for concept designs. Collected and summarized reports on timing closure and resource utilization.
- Developed C++ code for the Ultrascale+ hard processor to interface with custom RTL designed in Vitis Model Composer for loading and unloading image data between programmable logic and programmable software domains.
- Extensive practice with implementing critical ILAs for FPGA board debug and initial bring-up.

Firmware Development

- Developed Verilog RTL code to implement loopback for high-speed communication in electro-optic sensors using SERDES and UART protocols between FPGA and readout integrated circuits.
- Developed custom scripting language using C# to command-and-control special readout integrated circuits for highly sensitive Geiger Mode Avalanche Photo-Diode sensors (GmAPD).
- Designed UX/UI GUIs in C# to interface with equipment associated with Hardware in the Loop (HWIL) and Standardized Test Equipment (STEs).
- Wrote C# code to utilize dynamic link libraries to interface National Instruments test equipment to be controlled and monitored via custom graphical user interfaces.

Education

Master of Science in Electrical Engineering

December 2025

San José State University, GPA: 3.44

San Jose, CA

Emphasis: Artificial Intelligence & Digital Design

Bachelor of Science in Electronic Systems Engineering

May 2022

California State Polytechnic University, Pomona, GPA: 3.49

Pomona, CA

Emphasis: Digital System Design, Robotics, and Control

Graduate Projects

RapidLaneFPGA: FPGA Based Autonomous Car Highway Lane Segmentation System

- Developed a MobileNetV2 lane-segmentation CNN model and deployed it to FPGA using AMD Vitis AI and PyTorch. The Project utilized Deep Processing Unit (DPU) hardware on 224×224 images.
- Achieved 77.10% DICE accuracy on FPGA vs. 77.8% software baseline with INT8 quantization at 10 FPS (~100ms latency) on AMD ZCU104 Ultrascale+ device, demonstrating near-real-time inference.

FPGA Based LeNET Neural Network for Digit Recognition

- Implemented LeNet-based MNIST digit classifier using Vitis High Level Synthesis and deployed FPGA.
- Hardware inference output validated against software golden data and validated FPGA inference on PYNQ-Z1, successfully classifying digits 0–9.

Swimmers Pose Estimation Using Computer Vision and AI/ML

- Developed a computer vision pipeline for swimmer pose estimation using MediaPipe object detection, Segment Anything Model, and YOLOv7 to isolate swimmers in image and generated coordinates to train on a custom neural network.

FPGA Based Flight Controller, Search and Rescue Drone

- Developed an FPGA-based flight controller for a six-propeller search-and-rescue drone. The drone utilized a Zynq-7000 FPGA/SoC to process RC inputs and generate PWM motor control signals within the FPGA fabric.
- Designed a power distribution board for onboard camera and lighting systems.

Technical Skills

Vivado, Verilog, SystemVerilog, & VHDL

Vitis High Level Synthesis (HLS), Hardware Design using C++.

MATLAB/SIMULINK, Signal Processing, Control Systems Modeling, and FPGA RTL Design

QuestaSim, FPGA Hardware Simulation & Validation

Software Programming Languages, Working knowledge of Python and C/C++